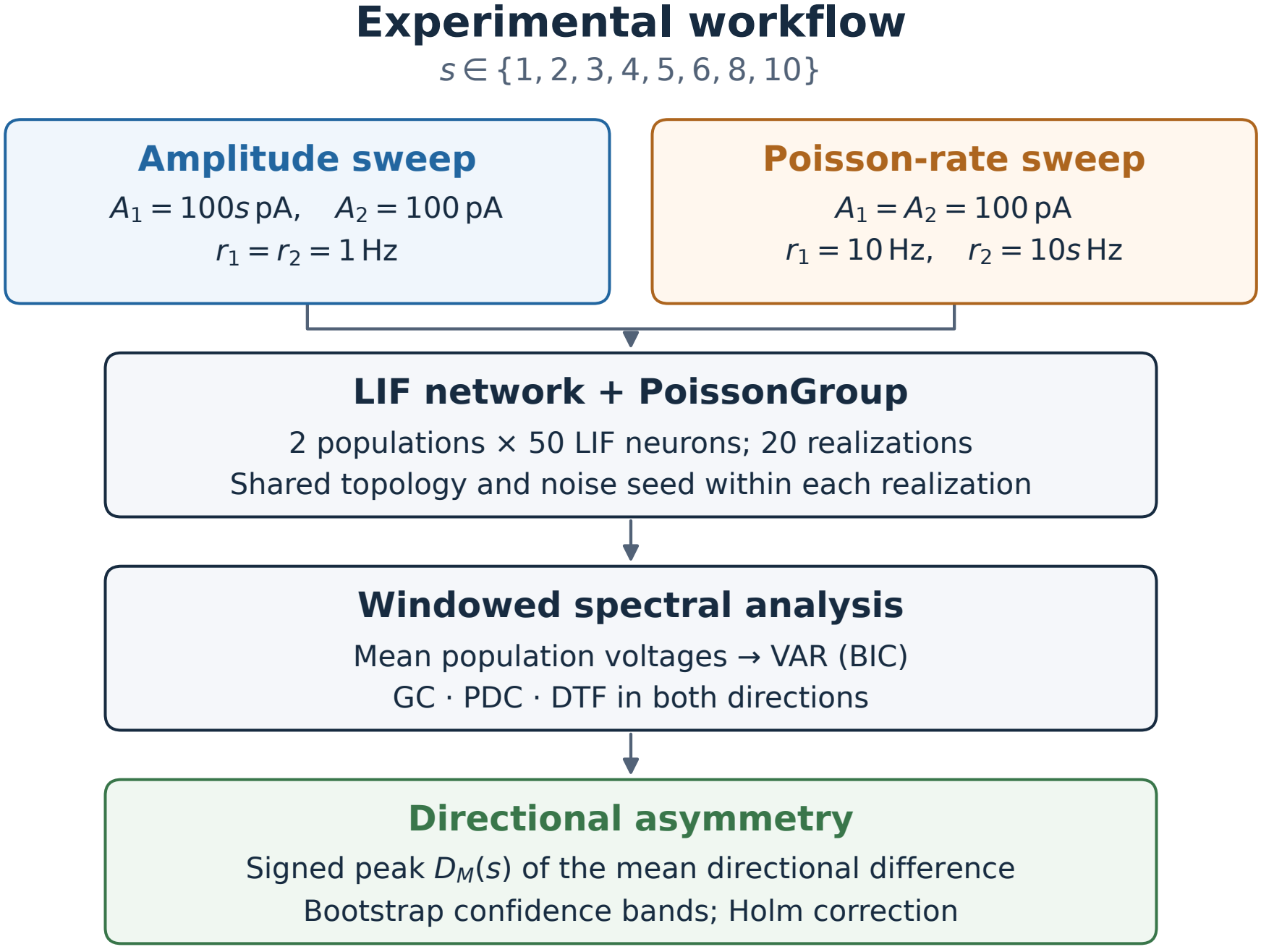
Experimental workflow
$s \in \{1, 2, 3, 4, 5, 6, 8, 10\}$
Amplitude sweep
$A_1 = 100s$ pA, $A_2 = 100$ pA
$r_1 = r_2 = 1$ Hz
Poisson-rate sweep
$A_1 = A_2 = 100$ pA
$r_1 = 10$ Hz, $r_2 = 10s$ Hz
LIF network + PoissonGroup
2 populations × 50 LIF neurons; 20 realizations
Shared topology and noise seed within each realization
Windowed spectral analysis
Mean population voltages → VAR (BIC)
GC · PDC · DTF in both directions
Directional asymmetry
Signed peak $D_M(s)$ of the mean directional difference
Bootstrap confidence bands; Holm correction